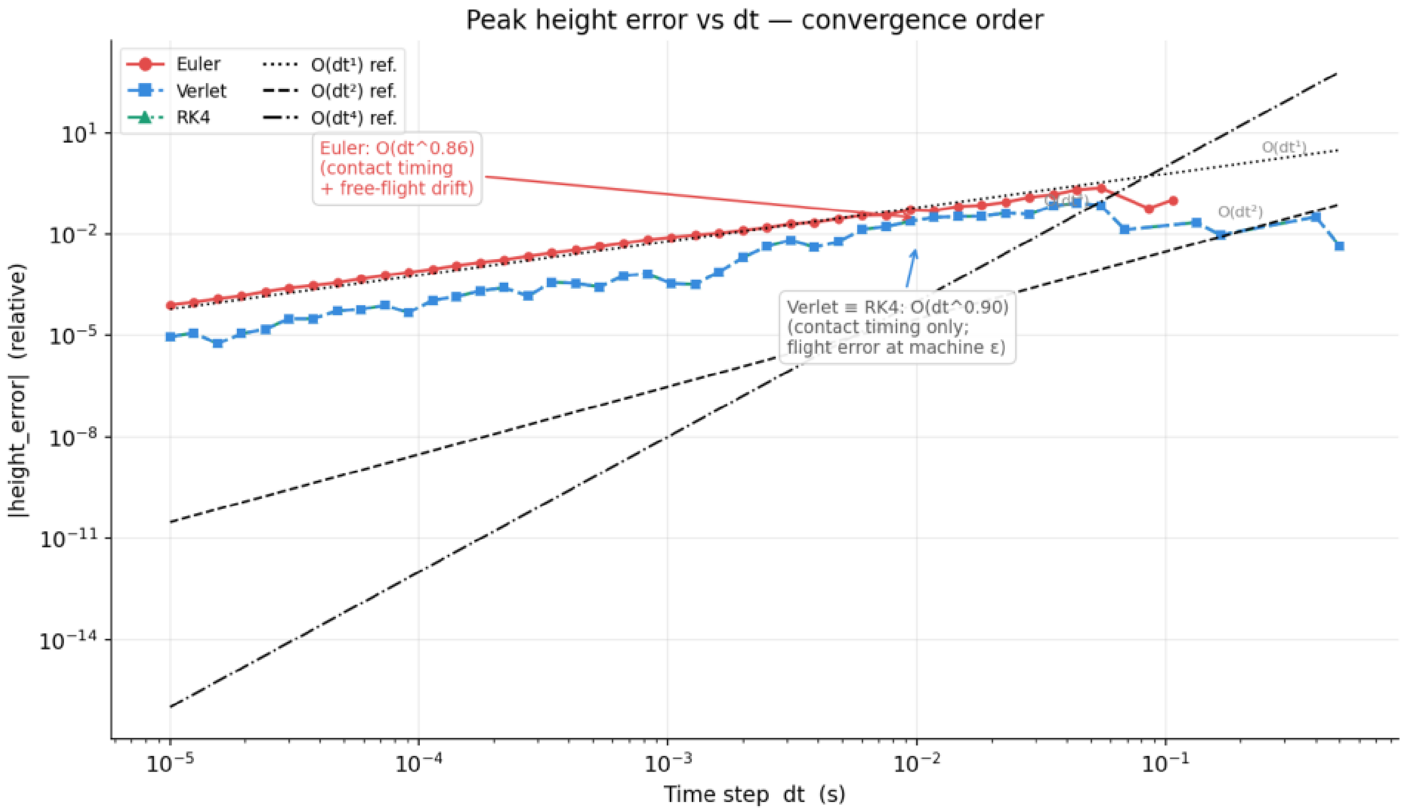


CONVERGENCE ORDER

Measured slopes (log-log fit):
- Euler → O(dt^0.86) R^2=0.978
- Verlet → O(dt^0.90) R^2=0.870
- RK4 → O(dt^0.90) R^2=0.870

All ~O(dt) – NOT O(dt), $O(dt^2)$, $O(dt^4)$ as theory predicts.

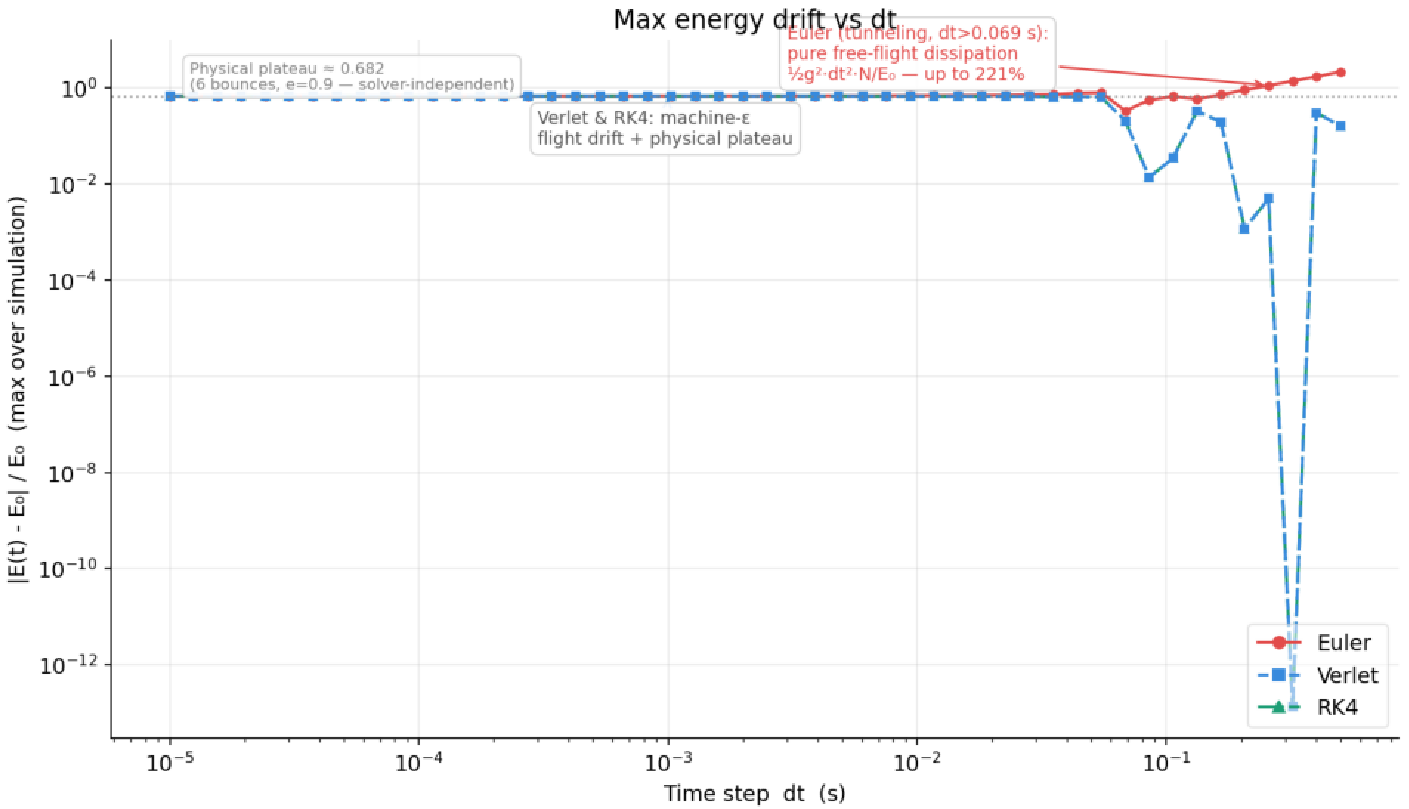
Root cause: discrete contact detection imposes an O(dt) floor on all integrators.

$v\_sim = v^* + g\cdot(dt-\tau) = v^*+O(dt)$
$\tau \in (0,dt)$: unknown contact frac.

Verlet ≡ RK4 (identical curves):
Free flight is quadratic → both exact → same state at contact → identical error.

Euler 8.6× higher than Verlet/RK4:
Extra $-\frac{1}{2}g^2dt^2$ dissipation/step.

→ Use Figure 2 to compare integrator quality in isolation.



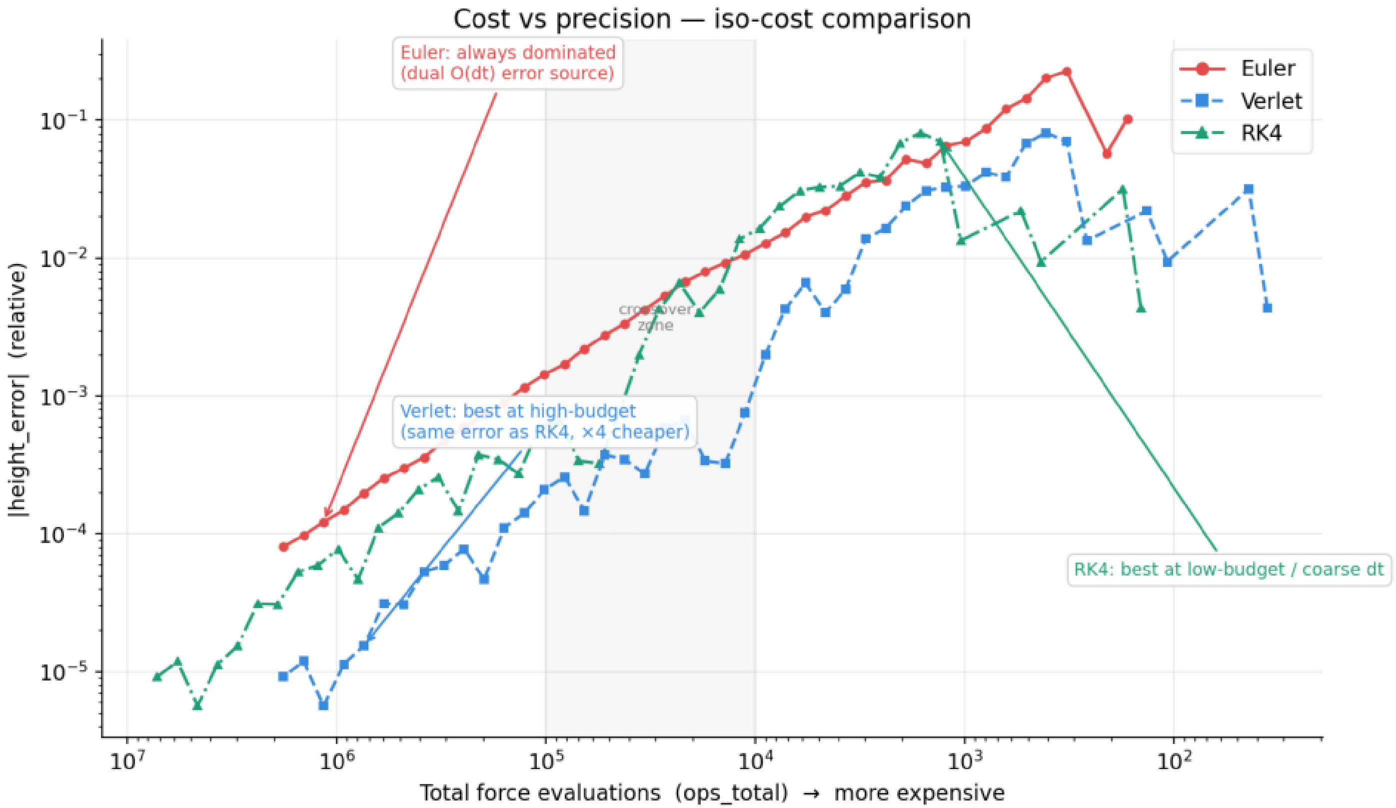
ENERGY CONSERVATION

Two distinct regimes:

Large dt — Euler only (dt>0.069 s):
Ball tunnels — no bounces.
Drift = pure Euler dissipation:
$\frac{1}{2}g^2 \cdot dt^2 \cdot N/E_0 \rightarrow$ up to 221%
(matches formula to 4 d.p.)
Verlet/RK4: machine ε (exact free-fall, no bounce losses).

Small dt — all solvers plateau:
max_energy_drift ≈ 68.2%
= physical_loss from 6 bounces with e=0.9 (analytical ≈ 64.7%; gap = contact timing impulse).

All three solvers reach the SAME plateau → energy floor is set by the impulse model, not the integrator.



COST vs PRECISION

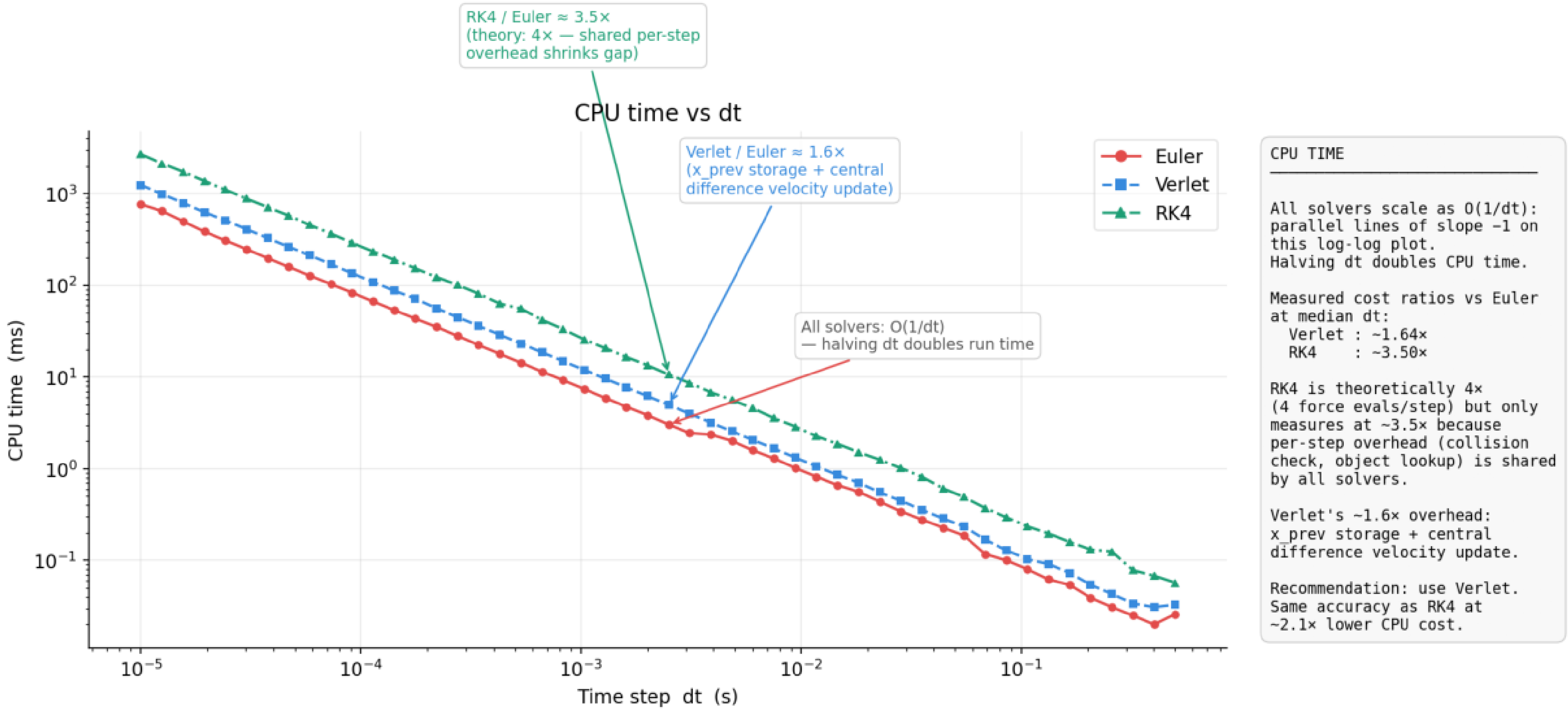
X-axis: total force evaluations
(Euler=1/step, Verlet=1/step,
RK4=4/step) ← left = costlier

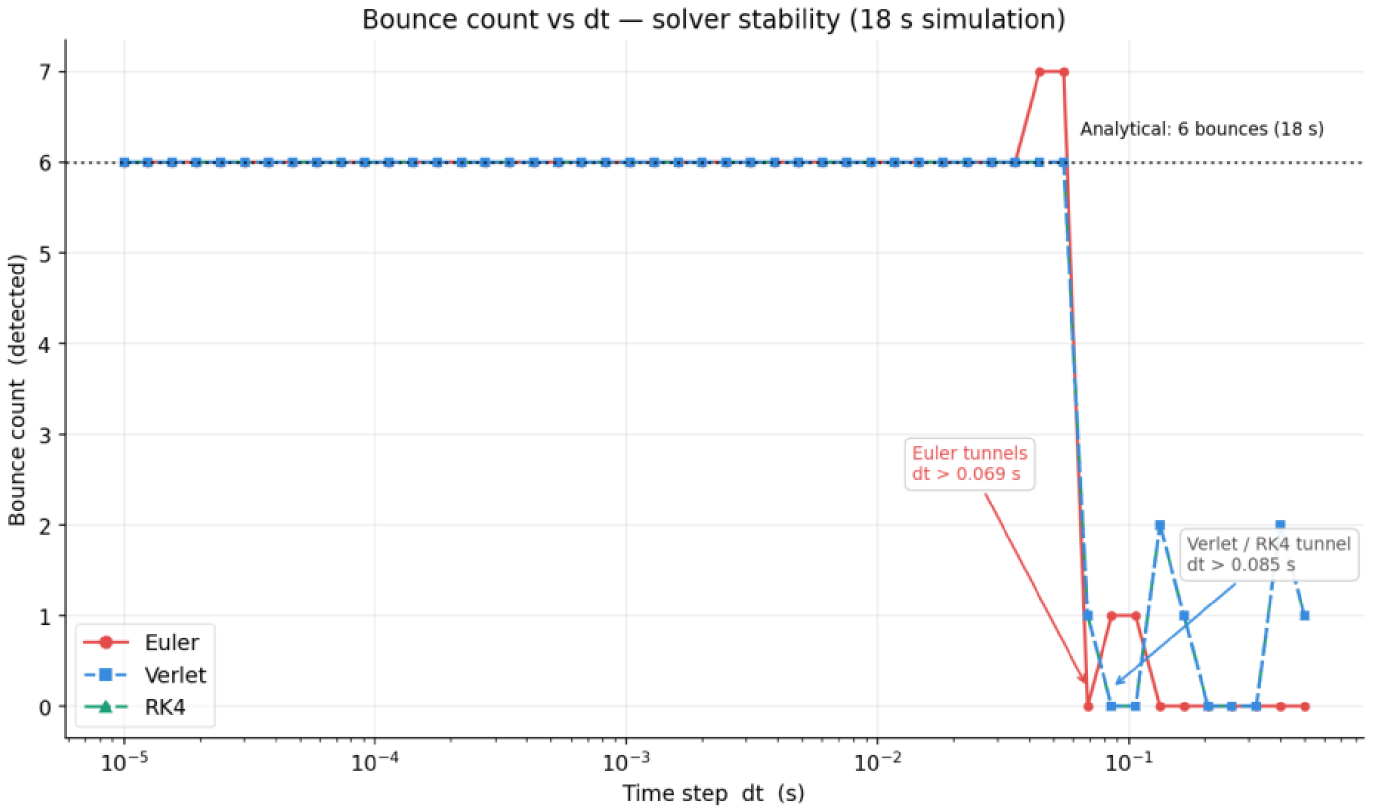
Euler: always dominated.
At any budget, Verlet or RK4
gives a much smaller error.

Verlet vs RK4:
Same O(dt) height error
(contact-timing-limited).
Verlet costs ¼ of RK4/step
→ identical precision at 4×
fewer ops → Verlet dominates.

At 60 Hz (120 ops/s, Verlet):
Verlet is the clear winner.

RK4 only wins at very coarse dt
(low-budget, rightmost region).
Never optimal with discrete
impulses — use CCD or spring
forces to expose $O(dt^4)$.


CPU time vs dt
RK4 / Euler ≈ 3.5×
(theory: 4× — shared per-step
overhead shrinks gap)
Verlet / Euler ≈ 1.6×
(x_prev storage + central
difference velocity update)
All solvers: O(1/dt)
— halving dt doubles run time
Euler
Verlet
RK4
CPU time (ms)
Time step dt (s)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
CPU TIME
All solvers scale as O(1/dt):
parallel lines of slope −1 on
this log-log plot.
Halving dt doubles CPU time.
Measured cost ratios vs Euler
at median dt:
Verlet : ~1.64×
RK4 : ~3.50×
RK4 is theoretically 4×
(4 force evals/step) but only
measures at ~3.5× because
per-step overhead (collision
check, object lookup) is shared
by all solvers.
Verlet's ~1.6× overhead:
x_prev storage + central
difference velocity update.
Recommendation: use Verlet.
Same accuracy as RK4 at
~2.1× lower CPU cost.



BOUNCE COUNT STABILITY

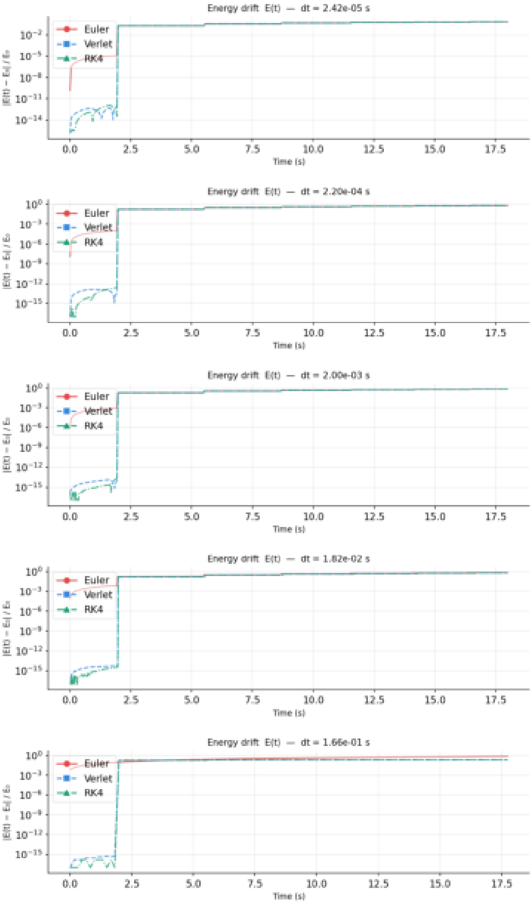
Analytical: 6 bounces in 18 s.
7th bounce at t≈18.07 s –
outside the window.

Zone 1 – Tunneling (large dt):
Euler : dt > 0.069 s → 0 bounces
Verlet/RK4: dt > 0.085 s
Both thresholds similar:
governed by approach velocity
at contact (v≈18.8 m/s), not
integrator order.

Zone 2 – Stable (small dt):
All solvers: 6 bounces.
Matches analytical exactly.
Euler has 7 bounces for 2 dt
values (residual timing offset).

Zone 3 – Euler transition:
Unstable (1–5 bounces) between
tunneling and stable zones.
Verlet/RK4: sharper transition.

E(t) time series — energy drift over simulation



ENERGY DRIFT E(t)

Each panel shows $|E(t)-E_0|/E_0$ over the full simulation. E_0 = initial kinetic + potential.

Staircase jumps = physical energy loss at each bounce ($1-e^2 \approx 19$ % per bounce, $e = 0.9$). Same for all solvers.

Between bounces (free flight):
- Euler → slow upward slope ($\sim\frac{1}{2}g^2dt^2$ loss per step, accumulates as $O(dt)$)
- Verlet → flat (exact for quadratic traj., machine ε)
- RK4 → flat (same reason)

Coarsest dt (1.66e-01 s):
Ball tunnels — no bounces. Only Euler linear drift is visible.

Time-domain view of the $O(dt)$ vs machine-ε gap quantified in Figure 2.